

AMRITA VISHWA VIDYAPEETHAM, AMRITAPURI

B. Tech CSE (ARTIFICIAL INTELLIGENCE)

LITERATURE REVIEW SUBMISSION

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4. Problem Definition:

Traditional Ayurvedic knowledge codified in Sanskrit texts such as *Charaka Samhitā*, *Suśruta Samhitā*, and *Bhāvaprakāśa* contains valuable medical insights but is expressed in an unstructured and linguistically complex form. Extracting structured knowledge from these digitized texts is challenging due to linguistic features such as Sandhi splitting, compounding, and free word order, as well as the absence of large, annotated corpora. While existing Sanskrit NLP resources are improving, they remain largely general-purpose and not tailored to the Ayurvedic domain. Hence, there is an urgent need for a domain-specific NLP pipeline that integrates morpho-syntactic analysis, Named Entity Recognition (NER), Relation Extraction (RE), and Knowledge Graph construction. This project aims to build such a pipeline using contemporary AI and transformer-based architectures, supported by ontology-driven annotation, to make Ayurvedic knowledge both easily queried and computationally accessible.

5. Literature survey Summary Table

Literature Survey Summary Table

No	Author(s) & Proceedings	Title	Year	Journal / Impact Factor	Citation	Dataset Used	Methodology / Approach	Limitations & Future works
1	Maheshwari et al., Findings of EMNLP	A Benchmark and Dataset for Post-OCR text correction in Sanskrit	2022	EMNLP Findings (ACL Anthology, IF ~3.4 equiv.)	~20	218k Sanskrit OCR-corrected sentences	Seq2seq correction baselines for Sanskrit OCR	Needs Ayurveda-specific adaptation. not domain-focused
2	Zhu et al., Frontiers in Genetics (HerbKG)	HerbKG: A Knowledge Graph of Herbal Medicine	2022	Frontiers in Genetics (IF 4.8)	~5	PubMed abstracts, herbal texts	BERT- based NER & RE for Herb- Chemical- Gene-Disease KG	Focused on Chinese medicine; adaptation needed for Ayurveda

3	Krishnan, Kulkarni, Huet, WSC	Digital Corpus of Sanskrit: Towards Improved Segmentation and Normalization	2023	World Sanskrit Conf.	~8	Digital Corpus of Sanskrit (DCS)	Normalization & segmentation improvements for Sanskrit	Lacks Ayurvedic corpora; calls for domain-specific expansion
4	Sandhan et al., ACL Demo (SanskritShala)	SanskritShala: A Neural Toolkit and Web Interface for Sanskrit NLP	2023	ACL Demo (ACL Anthology)	~4	Sanskrit corpora (segmentation, POS, parsing)	Neural toolkit for Sanskrit NLP tasks	Needs Ayurvedic lexicon & domain-specific tags
5	Terdalkar et al., arXiv / WSC	Ontology-Driven Knowledge Graphs for Ayurvedic Texts	2022	arXiv preprint, WSC 2023 paper	~6	Bhāvaprakāśanighaṇṭu (Dhānyavarga)	Ontology design + manual annotation + KG creation	Time-consuming manual process; limited scalability
6	Terdalkar et al., Āyurjñānam (WSC Poster)	Āyurjñānam: Queryable Knowledge Graph for Ayurvedic Texts	2023	WSC Poster / Demo	-	Annotated Bhāvaprakāśa KG	KG + query templates for Ayurvedic exploration	Limited automation; future work on auto-NER/RE
7	Nehrdich et al., Findings of EMNLP (ByT5-Sanskrit)	ByT5-Sanskrit: Multitask Pretraining for Sanskrit NLP	2024	EMNLP Findings (ACL Anthology)	~8	Multitask Sanskrit corpora	Byte-level T5 pretrained on Sanskrit, multitask fine-tuning	No Ayurvedic domain adaptation yet
8	Tandra (Sujoy) et al., WSC Workshop	Building Named Entity Resources for Sanskrit Texts	2023	World Sanskrit Conf. workshop	~7	Śrīmad-Bhāgavatam subset (NER corpus)	Pre-annotation + bilingual alignment + gazetteers	Entities limited to persons/places. Ayurveda entities missing
9	Trappey et al., MDPI Information	Ontology-Based Knowledge Graph Framework for Text Mining	2024	MDPI Information (IF 2.0)	~5	Patent + industrial corpora	Intelligent text mining + ontology-based KG framework	Generic; not made for ancient Sanskrit text
10	Behr et al., RSC Catalysis Sci. Technol.	Knowledge Graphs in Chemistry: Ontology Learning and NER Pipelines	2024	RSC Catalysis Sci. Technol. (IF ~5.7)	~7	Chemistry corpora	Review: NER + ontology learning + KG construction pipelines	Domain-agnostic; transferable methods only
11	Nedungadi et al., J. Ayurveda & Integrative Med.	Mapping Research Trends in Ayurveda: A Bibliometric Study (1993–2022)	2023	Elsevier JAIM (IF ~2.2)	~34	11,773 Ayurveda publications	Bibliometric mapping of Ayurveda research	Not technical NLP; shows domain significance
12	Bagchi et al., RSUSCI Conf.	NLP-Based Knowledge Extraction from Charak Samhita for Navigating Ancient Wisdom	2024	RSUSCI Conf 2024	-	OCR'd Charaka Samhitā	Django system: OCR + NER + query interface	Used translations, Sanskrit-native pipeline needed

13	Terdalkar & Bhattacharya, arXiv	Automated KG Creation and QA from Sanskrit Epics	2023	arXiv preprint	-	Mahābhārata, Rāmāyaṇa, Bhāvaṇaprakāśa synsets	Automated KG creation + QA in Sanskrit (includes Ayurveda terms)	Early-stage, needs full Ayurvedic ontology
14	Kulkarni et al., Sanskrit NLP Workshop	Advances in Morphological Segmentation and Compound Analysis in Sanskrit	2022	Sanskrit NLP Workshop	8	Sanskrit Digital Corpus	Morpho-syntactic parsing for Sanskrit	No Ayurvedic integration; useful for preprocessing
15	Kulkarni et al., WSC Proceedings	Sanskrit Semantic Parsing for Knowledge Extraction	2023	World Sanskrit Conf.	19	Sanskrit annotated datasets	Semantic parsing for extracting relations	Not domain-specific; Ayurveda corpus missing

6. Identified Research Gaps:

- **Lack of domain-specific annotated corpora:**

As there is a lack of large-scale annotated Ayurvedic corpus containing the type of entities like Dravya (herbs), Roga (diseases), Lakṣaṇa (symptoms), Doṣa etc., it hampers the development of precise Named Entity Recognition and Relation Extraction systems specific to Ayurveda.

- **Underexplored relation extraction for Ayurveda:**

Existing Ayurvedic Knowledge Graph efforts mainly annotate entities but rarely capture semantic relations such as “Herb–treats–Disease” or “Doṣa–causes–Symptom.” Automated RE in this domain is largely absent.

- **Lack of integration of contemporary Pre-trained Language Models with Ayurveda:**

Models such as ByT5-Sanskrit and SanskritShala perform well on segmentation and tagging but have neither been fine-tuned nor adapted on Ayurvedic material; thus, a gap remains to apply the latest NLP to Ayurveda.

- **Cross-script and transliteration difficulties:**

They were composed in Sanskrit/Devanāgarī but may be found transliterated into the Roman script in online resources. Free word order and sandhi splitting complicate consistent entity linking across the transliterations, making cross-lingual interoperability difficult.

7. Novelty / Contribution of the Proposed Work

We have identified multiple novelties that can be applied to this project. Depending on time and feasibility, we will be proceeding with one of them.

a) Creation of a curated Ayurvedic annotated corpus

- i) We will develop the first domain-specific annotated dataset for Ayurveda, labeling entities such as *Dravya* (herbs), *Roga* (diseases), *Lakṣaṇa* (symptoms), *Doṣa* (bio-energetic principles), and *Formulations*.
 - ii) Relation labels (e.g., *Treats*, *HasProperty*, *ContraindicatedFor*) will also be added, providing a resource not currently available in the field.
- b) Integration of advanced Sanskrit Pre-trained Language Models for Ayurveda**
- i) Fine-tuning byte-level pretrained models (e.g., ByT5-Sanskrit) for Named Entity Recognition and Relation Extraction in the Ayurvedic domain, enabling robust handling of Sandhi, compounding, and free word order.
 - ii) This represents one of the first applications of state-of-the-art Sanskrit PLMs directly on Ayurvedic corpora.
- c) Semi-automatic Knowledge Graph (KG) construction**
- i) Combining ontology-driven annotation (inspired by Sangrahaḥ/Āyurjñānam) with model-assisted pre-annotation and human-in-loop validation.
 - ii) The resulting Neo4j-based KG will allow query-driven exploration of Ayurvedic knowledge, bridging unstructured Sanskrit texts with structured medical data.
- d) Open-source reproducibility and evaluation**
- i) The entire pipeline (datasets, models, KG, and evaluation metrics such as F1-score, Labelled Attachment Score, and KG coverage) will be released in a reproducible form, ensuring future researchers and practitioners can build on this work.

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